AKI Project: Feasibility, Workload and Plan

Evaluation of the "Transportable early AKI prediction" project for a student group using MIMIC-IV, eICU-CRD and optionally SafeICU. The exact table and column details have not been verified against current documentation, so treat the specifics as a plan to confirm once data access is in place.

Verdict

AKI is a **medium-difficulty, medium-high effort** project. The modelling is the easy part. The hard parts are cohort and label construction, harmonizing two databases, and the evaluation design. It is doable for a group of 4-5 in about **10-12 weeks**, and a stripped-down version fits in about 8.

What exactly would be built

Task definition

This is the part that must be precise.

- **Population:** adult ICU stays (age 18 or older) with an ICU stay of at least 24h. Exclude end-stage renal disease or chronic dialysis, and any stay with renal replacement therapy before the prediction time.
- **Prediction points:** every 6h (or hourly) from ICU hour 12 onward. At each point, only patients **without AKI so far** are eligible.
- **Outcome:** new KDIGO AKI (stage 1 or higher) within the next **24h**. A 48h horizon and stage 2 or higher are secondary outcomes.
- **Label:** serum creatinine rise of at least 0.3 mg/dL within 48h, or at least 1.5 times baseline within 7 days. Baseline is the lowest creatinine in the preceding 7 days, or the admission value if there is no history. Urine output (below 0.5 mL/kg/h for 6h) is a secondary label only, because charting and weight data are inconsistent between datasets.
- **Features:** recent values plus min/max/mean/slope over 6/12/24h windows for creatinine, BUN, potassium, bicarbonate, lactate, MAP, heart rate, SpO2, urine output, vasopressor and diuretic use, nephrotoxic drugs, ventilation, age, sex and admission type. Also include "time since last measured" and missing indicators.

- **Missing data:** limited forward fill, indicator flags, and tree models that handle missingness natively.

Experiments

1. Baselines: logistic regression, then LightGBM/XGBoost.
2. Optional temporal model (GRU), only if it clearly beats LightGBM.
3. Internal validation: patient-level splits, plus a temporal split in MIMIC.
4. **Cross-dataset:** train MIMIC and test eICU, train eICU and test MIMIC, plus leave-hospital-out inside eICU.
5. Calibration under shift: reliability curves, Brier score, and recalibration with a small target-site sample.
6. Missing-data robustness: remove feature groups to simulate hospitals that measure less.
7. Explainability with SHAP, plus a check that explanations are clinically plausible.
8. Subgroups: age band, sex, baseline kidney function, ICU type.
9. Metrics: AUROC, AUPRC, calibration, and alert burden (alerts per patient-day at fixed sensitivity, and lead time before onset).

Application

A replay-based dashboard. Pick a patient, scrub through ICU time, and see the risk trend, calibrated probability, top drivers, alert threshold, and a data-completeness indicator. A Streamlit or FastAPI plus React front end works. It must be labelled a research prototype.

SafeICU

Treat it as optional. Pediatric creatinine norms are age-dependent, so adult-trained models should not be transferred as-is. Only include it if its documentation shows enough creatinine data and age strata to support a separate pediatric analysis. Otherwise state clearly why it was excluded.

Time and workload

Rough estimate for 4-5 people:

Phase	Weeks	Share of effort
Access, environment setup, clinical definitions, literature	1-2	10%
Data extraction, cohort and label construction (both datasets)	3-4	**35-40%**
Baselines, feature engineering, internal validation	2	15%
Cross-dataset, calibration, robustness, subgroups	2	15%
Application	2-3 (in parallel from week 4)	15%
Integration, evaluation freeze, report, presentation	2	10-15%

Expect roughly 25-35 person-weeks in total. Data engineering dominates, so do not underestimate it.

Requirements

- **Access:** PhysioNet credentialing (CITI training is already done) and signed data-use agreements for MIMIC-IV and eICU.
- **Skills:** SQL and pandas are essential. Someone needs to own cohort and label code, someone modelling and evaluation, someone the app, and someone the clinical definitions and writing.
- **Compute:** a modern laptop is enough for tree models on aggregated features. A GPU only matters if a GRU is used. Storage for MIMIC-IV and eICU is tens of GB once unpacked, so plan for a shared machine or a database (Postgres or DuckDB).
- **Time-critical first step:** do not wait for SafeICU to start. MIMIC and eICU are enough for the core project.

Main risks

1. **Label leakage and prevalent AKI.** The most common way this project goes wrong. Every feature must end at the prediction time.
2. **Informative lab ordering.** Creatinine is measured more often in sicker patients, so measurement frequency can leak the outcome.
3. **Harmonization.** Units, timestamps and variable names differ between MIMIC and eICU, so cross-dataset results can reflect mapping errors.
4. **Scope creep.** The sequence model, SafeICU and fancy app features are the first things to cut.

Minimum viable version: creatinine-based label, LightGBM vs logistic regression, MIMIC and eICU cross-validation, calibration, SHAP, and a simple replay dashboard. Everything else is a stretch goal.

Can Codex (or any AI coding agent) do it fully?

No, not as a hands-off job, and it should not be run that way.

- **What it can do well:** boilerplate and SQL, data-cleaning scripts, training pipelines, plotting, the API and front end, and tests. This can save a lot of time.
- **What it cannot do for you:**
  - Get credentialed and download the data.
  - Make clinical decisions, such as which KDIGO criteria to use, how to handle baseline creatinine, and which exclusions are defensible.
  - Catch subtle leakage or harmonization errors without human review of the logic. Agents often write code that runs fine but is scientifically wrong.
  - Defend the work in the viva. Every design choice must be understood by the team.
- **Data-governance warning:** PhysioNet's data use agreement for MIMIC and eICU restricts sharing the data with third-party online services, including AI tools, unless specific privacy conditions are met. Do not paste patient-level data or tables into any chatbot or cloud coding agent, and be careful about running an agent that can read the data directory. Use it on synthetic or schema-only examples, or check PhysioNet's current guidance first.
- **Academic integrity:** check what the course and supervisor allow for AI-assisted coding.

The sensible split is that the team owns the clinical definitions, cohort logic, validation design and interpretation, and uses an AI assistant for code speed-ups under review.

Next step

Provide team size, skills and deadline so this can be turned into an exact specification and a week-by-week plan.